and lipid metabolism when compared to insoluble fibres. Soluble fibre prolongs the rate of and gastric emptying and intestinal transit time. It forms a gel with water and thickens the unstirred layer. Carbohydrates are thus packed and insulated from the action of the digestive enzymes in the intestines thus reducing the rate of absorption. Insoluble fibre reduces gastric emptying and intestinal transit time. High fibre diet enhances the response of gastric inhibitory polypeptide (GIP) which is a stimulus for insulin secretion.



Activity : 5

List any five foods that are high in fibre and can be consumed by a diabetic.

1.

2.

3.

4.

5.

Always consume plenty of water at least 2 litres, if you are eating a fibre rich diet. Fibre without adequate fluids can lead to constipation. Increasing fibre intake slowly can also help to ease bloating or other unwanted gastrointestinal distress.

Protein

Protein should provide 12-20 % of energy intake. An additional 30 grams may be necessary during pregnancy and lactation. High protein intake helps to increase insulin production and promotes satiety. Protein requirements are increased in malnutrition, surgery or wound healing. Protein from vegetable source is preferable to that from animal sources.

Fat

The amount and type of fat plays an important role in the diet of a diabetic. Diabetics have a greater incidence of hyperlipidemia and atherosclerosis than do nondiabetics. The present recommendation of total fat is 20-30 % of total calories. Fat intake should be monitored carefully.

Saturated, monounsaturated and polyunsaturated fats are given in the ratio of 1:1:1.

Food Exchange lists

Food exchange lists are groups of measured foods of the same calorific value and similar protein, fat and carbohydrate and can be substituted for one another in a meal plan. The food exchange lists helps the patient to:

1. restrict the food intake according to the insulin prescription so that both hyperglycaemia and hypoglycaemia can be prevented.
2. have variety in the diet so that he can adhere to it always.
3. learn easily the principles of diet.

Dietary Guidelines

- Daily energy intake must be estimated after considering such fact as age, sex, activity and occupation.
- The total intake of calories is more important for a diabetic than the exact proportions of protein, fat and carbohydrate.
- A diabetic should maintain standard body weight or slightly lower.
- Simple sugars should be restricted since they are easily absorbed and have a high glycaemic index.
- Food exchange lists should be followed to prevent hypo and hyperglycaemia. The exchange lists should be followed to avoid monotony, dietary constancy and flexibility.
- Patients should avoid fasting and feasting.

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